Candidate's Name/Index Number……..

THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
013 BIOLOGY

Time: 2½ hours **November 2024**

INSTRUCTIONS

1. This paper consists of THREE (3) sections A, B, and C with a total of ten (10) questions.
2. Attempt **ALL** questions.
3. All answers must be written in the spaces provided.
4. Write your examination number on every page in this question paper.
5. All writing must be in blue/black ink. Diagrams must be drawn in pencil and labeled in ink.
6. Cell phones and calculators are not allowed in the examination room

SECTION A (15 MARKS)

1. Select the most correct answer in each item and write its letter beside its corresponding item number.
   (i) The next step after hypothesis formulation in the scientific procedure is known as
      (A) Data collection
      (B) Experimentation
      (C) Data analysis and interpretation
      (D) Asking questions

   (ii) One of the following is the most proper way of disposing waste:
      (A) Burying
      (B) Burning
      (C) Sanitary landfill
      (D) Open dumps

   (iii) One of the following is an example of an epidemic disease
      (A) Malaria
      (B) Cancer
      (C) Cholera
      (D) Heat attack

   (iv) When viewing objects under the microscope, a student at Seypalm Secondary school used X15 eye piece lens and X5 objective lenses. Find the total magnification.
      (A) X3
      (B) X75
      (C) X20
      (D) 150

Candidate's Name/Index Number……..

(v) Which of the following would be the first step in scientific research procedure?
- (A) Conclusion
- (B) Experimentation
- (C) Hypothesis formulation
- (D) Identification of the problem

(vi) Which among the following is a set of sexual transmitted disease?
- (A) HIV/AIDS, Cancer and gonorrhea
- (B) Syphilis, HIV/AIDS and gonorrhea
- (C) Tuberculosis, HIV/AIDS and Syphilis
- (D) Cancer, gonorrhea and syphilis

(vii) Which of the following would be the first line of defense as far as immunity is concerned?
- (A) White blood cells
- (B) Skin
- (C) Red blood cells
- (D) Tears

(viii) A condition in which a person has a difficult in breathing due to swallowing excess water is called:
- (A) Choking
- (B) Drowning
- (C) Fainting
- (D) Dying

(ix) Which of the following component of the first aid kit would you use in case of suffering a minor cut on your finger?
- (A) Elastic bandage
- (B) Liniment
- (C) Methylated spirit
- (D) Petroleum jelly

(x) What does one of the two main branches of Biology resents?
- (A) Scientific study of animals
- (B) Scientific study of matter in relation to energy
- (C) Scientific study of composition of matter
- (D) Scientific study of medicine

Answers:

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the phrase or term in list A with that in list B by writing its letter against the number of item in list A.

Candidate's Name/Index Number……..

LIST A		LIST B
(i)	It is located at the top of the body tube and it is used to magnify specimen after having been magnified by the objective lens.	A. Stage B. Mirror C. Ocular lens D. Coarse adjustment knob E. Condenser F. Fine adjustment knob G. Diaphragm H. Body tube
(ii)	A hollow tube that support the eyepiece and the revolving nose piece.	
(iii)	The place where the slide with specimen to be observed is placed.	
(iv)	It is located below the stage and is used to concentrate light on the specimen.	
(v)	It is used to regulate the amount of light passing through the condenser.	

i	ii	iii	iv	v

SECTION B (70 Marks)

Answer ALL questions from this section

3. When form one students were observing the environment with their biology teacher, they saw a lizard on a stone. Briefly explain any five characteristics that are observed in the lizard but are not observed in the stone.
   - ……
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   - ……
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   - ……
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   - ……
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   - ……
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4. Seypalm Secondary School is building a biology laboratory. As one of the brightest biology students at the school, advise them in ten points on the qualities the laboratory should have in order to be the best biology laboratory.
   - ……
   - ……
   - ……
   - ……
   - ……
   - ……
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   - ……
   - ……

- ……
- ……

5. In order to maintain good manners and proper personal hygiene, there are principles you can just follow. State any ten of the principles.

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6. Immunity can be categorized as Natural passive immunity, Natural active immunity, artificial passive immunity and artificial active immunity. Describe each of the above mentioned types of immunities.

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7. (a) Binominal nomenclature is a system of naming whereby each organism is assigned two names; the first name being the Genus and the second name, the species. Given that the genus of humans is 'homo' and the species is 'sapiens', explain the rules that must be adhered to when using binomial nomenclature to name organisms using humans for examples.

   (i)
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(ii)
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(iii)
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(b) Giving four points, briefly explain why artificial classification is not generally used when classifying organisms.

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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8. An attempt to grow maize on the Seypalm Secondary School farm failed after the maize wilted even though it was sufficiently watered. As one of the students in the school, you were charged with the task of making sure that next time you grow maize, it does not wilt again. Briefly explain the procedure you would follow to precisely find out why the maize wilted.

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9. Bacteria are organism of great economic importance because of their usefulness as well as some disadvantages to us. Outline eight ways that are useful and two ways that are not useful.
Usefulness:

(i) ……
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(ii) ……
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(iii) ……
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(iv) ……
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(v) ……
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(vi) ……
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(vii) ……
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(viii) ……
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Disadvantages:
(i) ……
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(ii) ……
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SECTION C (15 Marks)

10. Write an article that can educate the public about tuberculosis using the following guidelines:
    - Cause
    - Transmission
    - Symptoms/signs and effects
    - Control and treatment

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